
What predicts the next decade

CAPE, real rates, and other long-run forecasters tested across 145 years

ABSTRACT

We test the predictive power of cyclically-adjusted earnings yields (CAPE), real interest rates, and other valuation-based measures for U.S. equity returns over the subsequent ten years. Across 1,743 monthly observations from 1881 to 2025 with observable 10-year forward returns, the correlation between CAPE and forward 10-year real return is -0.511 , the strongest of the predictors examined. Sorting starting CAPE values into quintiles, the cheapest quintile (CAPE 4.8–11.1, mean 8.8) produces 10-year forward real returns averaging $+10.8\%$ annualised; the priciest quintile (CAPE 21.0–44.2, mean 26.2) produces $+3.1\%$. The current January 2026 CAPE of 30.8 places the U.S. equity market in the 95th percentile of the long-run distribution. We document the relationship and its limits, examine the post-1980 sub-sample where the predictive relationship has been weaker, and discuss what the elevated current valuation implies for forward expectations. We close with a discussion of the inherent limits of long-horizon return prediction and the appropriate uses of valuation-based forecasts.

KEY FINDINGS

- 01 CAPE-forward return correlation is -0.511 . Across 1,743 monthly observations 1881–2015 with observable 10-year forwards, the correlation is the strongest of the predictors examined. Stronger than dividend yield, real rate, or simple PE.
- 02 Quintile spread is 7.7 percentage points. Cheapest CAPE quintile (mean 8.8) $\rightarrow$ +10.8% real fwd 10y. Priciest (mean 26.2) $\rightarrow$ +3.1%. The relationship is monotonic and economically meaningful.
- 03 Current CAPE is at the 95th percentile. January 2026 CAPE of 30.8 places the market in the top 5% of the 145-year distribution. Implies forward 10-year real returns of approximately +1% to +3%.
- 04 Post-1980 the relationship has weakened. Correlation in the 1980–2015 sub-sample is -0.32 , materially lower than the full-sample -0.51 . The relationship is not dead but is less reliable in the recent era.
- 05 Out-of-sample R^2 is modest. Out-of-sample tests produce R^2 of 5–15% at 10-year horizons — meaningful but not decisive. CAPE is a probabilistic indicator, not a forecast.

Executive summary

Of the predictors of long-run equity returns that financial economics has produced, the cyclically-adjusted price-earnings ratio (CAPE) remains the most empirically reliable. The relationship is not strong enough to support market timing but is strong enough to materially update prior beliefs about the next decade.

This note tests the predictive power of CAPE, dividend yield, real interest rates, and several composite measures for U.S. equity returns over the subsequent ten years. Our sample covers 1881 through January 2026, with 1,743 monthly observations for which the 10-year forward return is observable (i.e., starting periods through January 2015).

The headline finding is that CAPE — the ratio of the current price to ten-year smoothed real earnings — exhibits the strongest predictive relationship. The correlation between log CAPE and subsequent 10-year real annualised return is -0.511 , statistically significant at the 0.001 level. The quintile spread is 7.7 percentage points: the cheapest quintile of CAPE starting points (mean CAPE of 8.8) produced subsequent 10-year real annualised returns averaging $+10.8\%$; the priciest quintile (mean CAPE of 26.2) produced $+3.1\%$. The relationship is monotonic across all five quintiles.

The current (January 2026) CAPE of 30.8 places the U.S. equity market in the 95th percentile of the long-run distribution. Only the late 1990s (CAPE peak above 44) and the 1929 peak (CAPE peak of 32) sat at higher levels. Applying the historical CAPE-forward-return relationship implies expected 10-year real returns of approximately $+1\%$ to $+3\%$, materially below the long-run mean of approximately $+6.5\%$.

Three caveats apply to the headline finding.

First, the relationship is probabilistic, not deterministic. Even at the 95th-percentile CAPE, individual 10-year forward outcomes have varied widely. The 1996 starting point (CAPE 28) produced subsequent 10-year real returns of approximately $+5\%$, somewhat above the regression line. The 2000 starting point (CAPE 43) produced approximately -4% , somewhat below. CAPE is informative about the central tendency of forward returns but not about the specific outcome.

Second, the post-1980 sub-sample shows a weaker relationship. Across 1980–2015 starting points, the CAPE-forward correlation is -0.32 , materially lower than the full-sample -0.51 . The relationship is not dead but it is less reliable in the recent era. Interpretations include: (a) structural change has reduced the predictive power, (b) the recent sample has too few CAPE peaks to estimate the relationship robustly, (c) the post-2008 zero-rate environment has shifted equilibrium valuations upward.

Third, out-of-sample tests produce modest R^2 : approximately 5–15% at 10-year horizons after accounting for the regression's lookback bias. CAPE is a meaningful but

not a decisive forecaster. Investors using CAPE to inform allocation should weight its signal accordingly — substantial enough to update priors, not large enough to support concentrated bets.

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What predicts the next

The note proceeds as follows. Section 2 reviews the literature. Section 3 presents the empirical analysis. Section 4 examines the post-1980 weakness. Section 5 discusses the current regime and concludes.

Introduction

The question of whether equity returns are predictable at long horizons is among the most-studied in financial economics. The case for predictability rests on the observation that valuations — measures of price relative to fundamentals such as earnings or dividends — exhibit substantial low-frequency variation, and that this variation is mean-reverting. If valuations are mean-reverting, then deviations from long-run averages must be corrected through some combination of price changes and fundamental changes. To the extent that price changes do the work, valuations are informative about subsequent returns.

The empirical history of this argument is decades long. [Campbell and Shiller \(1988\)](#) documented the relationship between dividend-price ratios and subsequent long-horizon returns, finding R^2 values of 30–40% at 10-year horizons. [Shiller \(2000\)](#) popularised the CAPE measure (price-to-trailing-10-year-real-earnings) and used it to issue prescient warnings about late-1990s valuations. [Asness \(2012\)](#) reinforced the empirical relationship and argued that CAPE remained informative despite repeated post-publication criticism.

The skeptical literature is equally substantial. [Welch and Goyal \(2008\)](#) conducted comprehensive out-of-sample tests of equity-return predictors and found that most lose much of their in-sample R^2 when tested out-of-sample. [Campbell and Thompson \(2008\)](#) responded by showing that simple constraints (no negative forecasts) restore meaningful out-of-sample power. [Siegel \(2016\)](#) argued that structural changes in accounting (notably the 2001 FAS 142/144 changes affecting earnings) make the post-2000 CAPE not directly comparable to historical values, and that adjusted CAPE measures are at lower percentiles than the raw figure suggests.

This note contributes to the literature with three specific aims. First, to provide an updated test of CAPE-forward predictability through the January 2026 endpoint, incorporating the post-COVID recovery and 2022 inflation episode. Second, to compare CAPE explicitly against alternative predictors (dividend yield, real rate, simple PE) on consistent methodology. Third, to apply the framework to the current January 2026 valuation and discuss the implied forward return expectation.

Methodology

2.1 Data and predictors

We use Shiller's monthly online series for prices, dividends, real earnings, CAPE (PE10), CPI, and long interest rates. The full sample runs from January 1871; we restrict to January 1881 onward because CAPE requires 10 years of preceding earnings data. The

10-year forward real return is computed as the geometric mean of monthly real total returns over the subsequent 120 months. The sample for which forward returns are observable runs through January 2015 (the last starting point with 10 full years of subsequent data through January 2025).

We test four primary predictors:

- **Log CAPE:** natural logarithm of Shiller's PE10. Logarithm is used because the relationship is approximately log-linear.
- **Dividend yield:** trailing dividend as a fraction of price.
- **Real interest rate:** long Treasury yield minus contemporaneous trailing 12-month CPI inflation.
- **Earnings yield (E/P):** inverse of trailing 12-month PE.

For each predictor, we report the unconditional correlation with subsequent 10-year real annualised return, the regression coefficient and intercept, the R² of the in-sample fit, and a quintile-based stratification.

2.2 Sample composition

TABLE 1. SAMPLE SUMMARY, JANUARY 1881 TO JANUARY 2015 STARTING POINTS

Statistic	Value
Number of monthly observations	1,609
CAPE range	4.8 to 44.2
CAPE mean	17.1
CAPE median	15.7
10y forward real return mean	+6.9%
10y forward real return median	+6.7%

03 Empirical results

3.1 Predictor comparison

TABLE 2. CORRELATION WITH SUBSEQUENT 10-YEAR REAL RETURN

Predictor	Correlation	In-sample R^2
Log CAPE	-0.511	26.1%
Earnings yield (E/P)	+0.482	23.2%
Dividend yield (D/P)	+0.435	18.9%
Real interest rate	+0.118	1.4%

Log CAPE produces the strongest unconditional correlation with forward 10-year real returns. Earnings yield (mathematically related to CAPE but using trailing 12-month rather than 10-year smoothed earnings) produces nearly as strong a correlation. Dividend yield is also strongly correlated, somewhat less so than CAPE. The real interest rate, by contrast, is weakly correlated, suggesting that the predictive content is in valuation rather than in macro variables per se.

The R^2 of 26% for log CAPE means that approximately one-quarter of the variation in 10-year forward real returns is explained by starting CAPE alone, in-sample. This is a substantial fraction by financial-economics standards; out-of-sample R^2 is typically smaller (5–15%) but remains meaningful.

3.2 CAPE quintile analysis

We sort the 1,609 starting observations into five equal-sized quintiles by CAPE and compute the realised 10-year forward real return distribution within each quintile.

TABLE 3. CAPE QUINTILES AND SUBSEQUENT 10-YEAR REAL RETURNS

Quintile	CAPE range	CAPE mean	Fwd 10y real (mean)	Fwd 10y real (median)
Q1 (cheapest)	4.8 – 11.1	8.8	+10.79%	+10.91%
Q2	11.1 – 14.4	12.6	+7.16%	+7.04%

Quintile	CAPE range	CAPE mean	Fwd 10y real (mean)	Fwd 10y real (median)
Q3	14.4 – 17.3	15.9	+6.51%	+6.83%
Q4	17.4 – 21.0	19.0	+5.66%	+5.46%
Q5 (priciest)	21.0 – 44.2	26.2	+3.09%	+3.50%

Q1 FWD REAL

+10.79%

Q5 FWD REAL

+3.09%

QUINTILE SPREAD

7.7 pp

The quintile pattern is monotonic: each successive quintile of CAPE produces lower forward 10-year real returns. The spread between extremes is 7.7 percentage points — a substantial economic magnitude that compounds over a 10-year horizon to a 2× wealth difference. Investors starting at Q1 CAPE values experienced double-digit real returns over their next decade; those starting at Q5 values experienced low single-digit returns.

3.3 Distribution within quintiles

The mean and median statistics conceal substantial within-quintile variation. Even within the cheapest quintile, individual 10-year forward outcomes range from approximately +3% to +18% real annualised. Within the priciest quintile, outcomes range from approximately –5% to +9%.

TABLE 4. DISTRIBUTION OF FORWARD RETURNS WITHIN CAPE QUINTILES

Quintile	10th percentile	90th percentile
Q1	+3.7%	+17.5%
Q2	+1.9%	+12.4%
Q3	+1.2%	+11.0%
Q4	–0.6%	+11.4%
Q5	–4.5%	+9.1%

The 80% confidence intervals (10th to 90th percentile) within each quintile span 10–14 percentage points. The quintile-mean spread of 7.7pp between extremes is therefore smaller than the within-quintile noise. CAPE shifts the central tendency but does not pin down the specific outcome.

This is the appropriate way to read the predictive relationship. CAPE is informative about the distribution from which the next decade's returns are drawn; it is not a forecast of the specific realised value. An investor using CAPE-based estimates should treat them as central tendencies with substantial confidence intervals, not as point forecasts.

The post-1980 weakness and current valuation

4.1 Sub-sample evidence

A natural concern with valuation-based predictors is that the empirical relationship may have shifted over time. Specifically, the 1980–2010 period was characterised by structural disinflation, falling real interest rates, and PE expansion that may have shifted the equilibrium relationship between CAPE and subsequent returns.

TABLE 5. CAPE-FORWARD CORRELATION BY SUB-SAMPLE

Sub-sample	Observations	CAPE-fwd corr.	In-sample R ²
1881–1980 starting points	1,189	−0.572	32.7%
1980–2015 starting points	420	−0.319	10.2%
Full sample	1,609	−0.511	26.1%

The post-1980 correlation is materially lower than the pre-1980 correlation: −0.32 versus −0.57. The R² in the post-1980 sub-sample is approximately one-third of the pre-1980 figure. The relationship is not dead — it is still negative and statistically significant — but it is less reliable in the recent era.

Several explanations are plausible. [Siegel \(2016\)](#) argues that accounting changes (FAS 142 in 2001 and FAS 144 in 2002) have reduced post-2000 reported earnings relative to economic earnings, causing CAPE to overstate "true" valuations. [Philosophical Economics \(2013\)](#) and others argue that secularly higher payout ratios via buybacks have shifted the appropriate valuation benchmark. [Asness \(2012\)](#) argues that the post-1980 sample contains too few CAPE peaks to estimate the high-CAPE end of the distribution robustly, but that the relationship is not structurally different.

The interpretation matters for current forecasting. If the relationship is structurally weaker, then high CAPE today implies less negative implications for forward returns than the historical relationship suggests. If the relationship is just noisily estimated post-1980, then the historical relationship remains the appropriate guide.

4.2 The current January 2026 valuation

As of January 2026, the S&P 500 CAPE stands at approximately 30.8. The historical distribution places this:

- **Above the long-run mean of 17.1** by approximately 80%.
- **Above the long-run median of 15.7** by approximately 95%.

- **At the 95th percentile** of the 145-year monthly distribution. Only the late 1990s peak (CAPE above 40) and a brief 1929 peak (CAPE 32) exceeded the current level.

Applying the full-sample CAPE-forward regression directly produces an expected 10-year forward real return of approximately +1.5% from current valuations. Applying the post-1980-only relationship produces approximately +3% (the relationship is weaker, so the implied return is closer to the unconditional mean). The point estimate from the full sample is therefore the more pessimistic; from the recent sample, modestly less so.

An investor needs to take a position on which sample is the appropriate guide. Our view is that neither sample is decisively correct; the appropriate answer is some weighted average. A reasonable weighted average — for instance, equal weighting — produces an implied forward 10-year real return of approximately +2% to +3%, materially below the long-run unconditional mean of +6.7%.

TABLE 6. IMPLIED FORWARD RETURNS AT CURRENT CAPE

Methodology	Implied 10y real ann.
Full-sample regression	+1.5%
Post-1980 regression	+3.0%
Bayesian average	+2.2%
Quintile (Q5) historical mean	+3.1%
Long-run unconditional mean	+6.7%

4.3 What this implies for portfolio construction

An expected forward return of 2–3% real on U.S. equities does not imply that equities should be abandoned. It implies that:

- Withdrawal-rate planning calibrated against the long-run 6.7% real should be revised downward.
- The expected premium of equities over bonds is compressed, since bond yields are at or above their long-run norm.
- Diversification into less-overvalued markets (international developed, emerging) and into assets with regime-conditional behaviour (commodities, gold, real assets) is more valuable when the home market is at the 95th percentile of its valuation distribution.
- Investors near retirement, with shorter horizons over which the historical relationship is even noisier, should weight valuation considerations more heavily than younger investors with longer horizons over which the relationship is more reliable.

The CAPE signal is not a sell signal. It is a calibration signal — informing what level of forward returns is reasonable to expect, conditional on the historical record holding approximately as it has.

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Discussion, limitations, and conclusion

Limitations. Our analysis is U.S.-only. Cross-country tests ([Asness, Ilmanen, and Richardson, 2017](#)) find similar relationships internationally but with substantial cross-country variation. The CAPE measure has been criticised on accounting grounds (Siegel, 2016) and on payout-ratio grounds (Philosophical Economics, 2013); we report results using raw CAPE without these adjustments, as the adjusted measures are themselves contested and add specification uncertainty. Our forward-return horizon is fixed at 10 years; shorter horizons produce weaker relationships, longer horizons stronger ones, but the 10-year choice matches most academic literature.

What CAPE is not. CAPE is not a market-timing signal. The 1996 starting point (CAPE 28) preceded four more years of strong returns and a 1999 CAPE peak above 40 before the 2000 reversal. An investor who exited equities in 1996 on CAPE grounds underperformed substantially through 1999. CAPE provides probabilistic information about long-horizon central tendencies, not signals about timing.

Conclusion. CAPE is the most empirically reliable predictor of long-horizon U.S. equity returns. The full-sample correlation of -0.51 is robust, monotonic in quintile sorting, and economically meaningful with a quintile spread of 7.7 percentage points in expected 10-year real returns. The post-1980 sub-sample shows a weaker but still negative relationship. The current January 2026 CAPE of 30.8 places the U.S. equity market at the 95th percentile of its 145-year valuation distribution, implying expected forward 10-year real returns of approximately 2–3% — meaningfully below the long-run unconditional mean of 6.7%. The implications for portfolio construction are calibration-related rather than timing-related: investors should adjust forward expectations downward but should not necessarily abandon equity exposure.

CAPE does not **predict** the next decade. It **conditions** it. The current 95th-percentile reading does not imply a crash — it implies an unusually narrow range of likely outcomes, with a central tendency materially below the historical mean.

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Appendix A: Robustness checks

A.1 Alternative valuation measures

TABLE A1. ALTERNATIVE VALUATION PREDICTORS

Measure	Correlation with fwd 10y real
Log CAPE	−0.511
Earnings yield (E/P)	+0.482
Dividend yield (D/P)	+0.435
Cyclically-adjusted dividend yield	+0.487
Tobin's Q (proxy)	−0.398
Market cap / GDP	−0.421
Q5/Q1 implied 10y spread	7.7 pp (CAPE basis)

All valuation measures show meaningful predictive power, with CAPE marginally the strongest. Combining predictors via principal-components analysis produces an aggregated indicator with correlation approximately −0.55, modestly improved from CAPE alone but with substantially more specification complexity.

A.2 Out-of-sample tests

We perform a rolling out-of-sample test, estimating the CAPE regression on observations preceding each forecast date and producing forecasts that use only past information. The out-of-sample R^2 is computed against the realised 10-year forward return:

TABLE A2. OUT-OF-SAMPLE R^2 BY TRAINING WINDOW

Training window	Out-of-sample R^2
30 years	5.2%
50 years	8.7%
80 years	14.1%
Full prior history	15.3%

Out-of-sample R^2 is materially lower than in-sample R^2 (which was 26%) but is meaningful and positive across training windows. The R^2 of 14–15% is substantially better than the unconditional mean as a forecast and is consistent with the academic-literature consensus on CAPE-based predictability.

A.3 The 1996 false alarm

The most-cited "false alarm" example for CAPE is the 1996 starting point. CAPE crossed 28 in late 1996, prompting [Shiller's \(2000\)](#) warnings. Subsequent 10-year real returns from late 1996 averaged approximately +5% — modestly ab

A.4 ProRealTime: CAPE-conditional regime filter

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// CAPE-conditional long-only equity exposure
// Daily bars, CAPE updated monthly (manual feed or screener)

DEFPARAM CumulateOrders = False

// CAPE bucket thresholds from the 155-year sample
cape_low  = 13
cape_high = 28

// In practice, cape value is loaded as a custom data series
currentCAPE = customClose

// Allocation rule based on CAPE percentile bucket
IF currentCAPE < cape_low THEN
    targetExposure = 1.5 // overweight when cheap (leverage permitted)
ELSIF currentCAPE > cape_high THEN
    targetExposure = 0.3 // underweight when rich
ELSE
    targetExposure = 1.0
ENDIF
```

```
// Translate to position sizing via your normal sizing engine
// See companion paper on position sizing (Mar 2020)
```

ove the regression-implied value but still below the unconditional mean. CAPE was directionally correct (high CAPE → below-average returns) but not perfectly calibrated. The episode illustrates that CAPE provides probabilistic information rather than precise forecasts; a directionally correct signal can still produce return realisations within 2–3 standard deviations of either side of the implied central tendency.

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